

Crowd Size Analysis Report

Lexington Battle Green Event

Date: October 18, 2025
Time: Approximately 10:30 AM
Location: Lexington Battle Green, Lexington, Massachusetts

Executive Summary

This report provides an analysis of crowd size at the Lexington Battle Green based on two aerial photographs taken on October 18, 2025, at approximately 10:30 AM. Using established crowd estimation methodologies and aerial density analysis, the estimated crowd size is **8,000-12,000 attendees**.

Aerial Documentation

Photo 1: Primary Gathering Area



Overhead view showing the main concentration of attendees with clear visibility of the stage/speaking area and surrounding crowd density patterns.

Photo 2: Extended View



Alternative aerial perspective showing the full extent of the gathering, including peripheral areas and access points along adjacent streets.

Methodology

1. Area Measurement

Using the known dimensions of Lexington Battle Green (approximately 2 acres or 8,000 square meters), the crowd coverage area was estimated from the aerial photographs:

- **Primary gathering area:** Approximately 5,000-7,000 square meters
- **Pattern:** Dense concentration near central stage area, with decreasing density toward perimeter

2. Density Classification

The crowd was analyzed using standard density classifications for outdoor gatherings:

Zone Type	Density Range	People per m ²
Dense Core	Near stage/focal point	2-3 per m ²
Moderate	Mid-range areas	1-1.5 per m ²
Peripheral	Outer edges	0.5-1 per m ²
Sparse	Far perimeter	0.3-0.5 per m ²

3. Calculation Approach

Conservative Estimate: - Dense core ($\approx 1,500\text{ m}^2$) $\times 2.5\text{ people/m}^2 = 3,750$ - Moderate zones ($\approx 2,500\text{ m}^2$) $\times 1.25\text{ people/m}^2 = 3,125$ - Peripheral areas ($\approx 1,500\text{ m}^2$) $\times 0.75\text{ people/m}^2 = 1,125$ - **Subtotal: $\approx 8,000$ people**

Upper Estimate: - Dense core ($\approx 2,000\text{ m}^2$) $\times 3\text{ people/m}^2 = 6,000$ - Moderate zones ($\approx 3,000\text{ m}^2$) $\times 1.5\text{ people/m}^2 = 4,500$ - Peripheral areas ($\approx 2,000\text{ m}^2$) $\times 1\text{ people/m}^2 = 2,000$ - **Subtotal: $\approx 12,000$ people**

Analysis and Observations

Spatial Distribution

- **Central concentration:** Highest density observed in the area immediately in front of the stage/speaking platform
- **Radial dispersion:** Crowd density decreases progressively moving away from the focal point
- **Tree coverage:** Multiple mature trees create natural breaks in crowd formation
- **Access points:** Visible gathering along walkways and street edges

Visual Indicators

- Clear evidence of organized event with staging infrastructure
- Multiple signs and banners visible throughout the crowd
- Organized standing formation suggesting planned gathering rather than spontaneous assembly
- Some attendees visible on adjacent streets and pathways

Limitations

1. **Tree canopy coverage:** Portions of the crowd are obscured by foliage, potentially undercounting attendees
2. **Shadow areas:** Long shadows in the morning light reduce visibility in certain sections
3. **Movement:** As a snapshot in time, the count does not account for arrivals or departures
4. **Peripheral areas:** Difficulty distinguishing between active participants and passersby in outer zones

Conclusions

Primary Finding

Based on aerial photographic analysis, the crowd size at Lexington Battle Green on October 18, 2025, at approximately 10:30 AM is estimated at **8,000-12,000 attendees**.

Best Estimate

The most probable attendance figure is **approximately 10,000 people**, representing the midpoint of the estimated range and accounting for both visible and partially obscured areas.

Significance

This represents a substantial public gathering for the Lexington Battle Green venue, which has a total area of approximately 2 acres. The density and organization suggest a well-attended planned event.

Confidence Level

- **High confidence:** 8,000-12,000 range ($\pm 20\%$ margin)
- **Moderate confidence:** 10,000 as point estimate
- **Variables affecting accuracy:** Tree coverage, shadow interference, perimeter ambiguity

Technical Notes

Analysis Date: October 18, 2025
Analyst Method: Aerial density analysis with grid-based estimation
Reference Standards: Industry standard outdoor crowd density metrics
Image Quality: High-resolution aerial photography with clear visibility

This report provides a professional estimate based on available photographic evidence. For official capacity planning or permitting purposes, ground-based counting methods or electronic monitoring may provide additional precision.